Fiscal Year:	FY 2009	**Task Last Updated:**	FY 09/25/2012
PI Name:	Wayner Jr., Peter Ph.D.		
Project Title:	R & D work on the Constrained Vapor Bubble System for a Microgravity Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Interfacial phenomena		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wayner@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6199
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Isermann Department of Chemical Engineering		
PI Address 2:	110 8th Street		
PI Web Page:			
City:	Troy	**State:**	NY
Zip Code:	12180	**Congressional District:**	21
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	04/05/2005	**End Date:**	06/01/2009
No. of Post Docs:		**No. of PhD Degrees:**	2
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Plawsky, Joel (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNC05GA27G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE THIS IS A CONTINUATION OF FUNDING PROPOSAL FOR NAG3-2383. The thermo-physical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, the authors proposed the experimental and theoretical study of the Constrained Vapor Bubble Heat Exchanger (CVB). It is a flight development project. Results obtained on the ISS under micro-gravity conditions will be compared with ground-based results obtained at Rensselaer. The dimensions of the passive CVB heat exchangers were optimized for use under micro-gravity conditions where enhanced heat transfer should be achieved. The unique CVB system allows multi-faceted research and development: 1) it is a study of a passive heat exchanger optimized for micro-gravity; 2) it is a basic engineering study in thermal transport; and 3) it is a basic scientific study in interfacial phenomena, microgravity fluid physics and thermodynamics. The results will be used to design improved passive heat exchangers, which are needed for space exploration.		

Task Description:	Ground-based research and development work to support our flight experiment on the International Space Station was proposed for the five-year period starting April 22, 2000. Research on special experimental and modeling techniques was proposed. Assistance with the development work for the flight experiment itself at the NASA Glenn Research Center (and the external contractor Northrop-Grumman / Zin) along with necessary basic ground-based research and post flight analyses of the data were envisioned. The proposed work was needed to accomplish the science requirements and test matrix outlined in our revised Science Requirements Document (60055-DOC-000) dated October 1999, and to extend the fundamental understanding of transport processes controlled by interfacial phenomena. The latest revision (F) of the SRD is dated July 4, 2004. We are working with a NASA Projects Team headed by Sue Motil, and technical monitors R. Balasubramaniam and Sang Young Sonat at the NASA Glenn Research Center on the design of a microgravity experimental system for the International Space Station. It is anticipated that the flight experiment will be flown in the Fluids Integrated Rack, FIR, using the Light Microscope Module, LMM, on the International Space Station in the Year 2007. Unique low capillary pressure systems for high heat fluxes under microgravity conditions are being emphasized.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	

Work was completed on the design of the microgravity experimental system for the International Space Station. Results documenting these studies were either published or accepted for publication. The flight experiment on the ISS was completed under the subsequent Grant/Contract # NNX09AL98G (PI=Plawsky_.
The following presentations were made:

PRESENTATIONS [PR]:

PR 01) Presented at 2005 Annual AIChE Meeting (Cincinnati, OH, Oct 30 – Nov. 4) AICHE Paper #22828 : Sashidhar S. Panchamgam, Joel L. Plawsky and Peter C. Wayner, Jr. “Spreading Characteristics and Microscale Evaporative Heat Transfer in a Moving Meniscus Containing a Binary Mixture”

PR02) Presented at 2005 Annual AIChE Meeting (Cincinnati, OH, Oct 30 – Nov. 4) AICHE Paper #23015 : Sashidhar S. Panchamgam, Joel L. Plawsky and Peter C. Wayner, Jr. “Microscale Heat Transfer and Fluid Flow in an Evaporating Moving Extended Meniscus”

PR 03) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr.,”Microscale Heat Transfer in an Evaporating Moving Extended Meniscus”, ECI Conference on Microscale Heat Transfer, Sept. 25-30, 2005, Castlevecchio Pascoli (Tuscany) Italy.

PR 04) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr., ”Reflectivity Based Analysis of Contact Line Behavior and Microscale Heat Transfer in Ultrathin Films,” Poster at the 2nd Annual Tech Valley Engineering Symposium, April 18, 2006, Albany, NY.

PR 05) “Influence of Marangoni Stresses and Slip on Spreading Characteristics of an Evaporating Binary Mixture Meniscus” Presented at 9th Joint AIAA/ASME Thermophysics and Heat Transfer Conference, San Francisco, CA June 5-8, 2006: Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr.

PR 06) Presented at 4th International Conference of Nanochannels, Microchannels and Minichannels, June 19-21, 2006, Stokes Research Institute-University of Limerick, Limerick, Ireland: .Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr. , “Experimental Evaluation of Marangoni Shear in the Contact Line Region of an Evaporating 99 + % Pure Octane Meniscus.”

PR 07) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr., "Reflectivity Based Analysis of Contact Line Behavior and Microscale Heat Transfer in Binary Ultrathin Films,” AIChE 2006 Annual Meeting, November 12-17, 2006, San Francisco, California.

PR 08) Peter C. Wayner, Jr., “Effects of Interfacial Phenomena and Conduction on an Evaporating Meniscus,” ICNMM2007-30021 Fifth International Conference on Nanochannels, Microchannels and Minichannels, June 18-20, 2007. Universidad De Las Americas Puebla, Mexico.

PR 09) Paper 304b Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner,Jr., “A Multiscale Model For The Constrained Vapor Bubble Heat Pipe” presented at 2007 Annual AIChE Meeting, November 4-9, Salt Lake City, Utah.

PR 10) Paper # 2008-0818. 22nd Symposium on Gravity-Related Phenomena in Space Exploration, January 7-10, 2008, Reno, NV. “R&D Work on the Constrained Vapor Bubble System for a Microgravity Experiment”, Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr.

PR11) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. “ Continuum Models with Slip for the Evaporating Meniscus and Comparison with Experimental Results” Paper Number: CNMM2008-62080 Sixth International Conference on Nanochannels, Microchannels and Minichannels,”, TU Darmstadt, Germany, June 23-25, 2008.

PR12) Keynote Presentation: Peter C. Wayner, Jr., “Thermal-Physical-Chemical Phenomena in the Contact Line Region”, The Third International Topical Team Workshop on Two-Phase Systems for Ground and Space Applications in Brussels, Belgium September 10-12, 2008.

PR13) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. "Oscillations In An Evaporating Meniscus" ID#: 126554, Presented in Engineering Sciences and Fundamentals Session, 2008 Annual Meeting of AIChE, November 16-21, 2008 Philadelphia, PA.

PUBLICATIONS [PU] are listed below and in the Bibliography section:

PU 01) S. J. Gokhale, J. L. Plawsky, and P.C. Wayner, Jr. , “Spreading, Evaporation, and Contact Line Dynamics of Surfactant Laden Micro-Drops” Langmuir, Vol. 21, pp 8188-8197, No. 18, Sept. 2005.

PU 02) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr., ”Microscale Heat Transfer in an Evaporating Moving Extended Meniscus”, Experimental Thermal and Fluid Sciences, Vol. 30, pp 745-754 (2006),

Task Progress:	PU 03) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr. , "Influence of Marangoni Stresses and Slip on Spreading Characteristics of an Evaporating Binary Mixture Meniscus" Proceedings of the 9th Joint AIAA/ASME Thermophysics and Heat Transfer Conference, San Francisco, CA June 5-8, 2006. PU 04) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr. , "Experimental Evaluation of Marangoni Shear in the Contact Line Region of an Evaporating 99 + % Pure Octane Meniscus" Paper # ICNMM2006-96060, Proceedings of 4th International Conference of Nanochannels, Microchannels and Minichannels, June 19-21, 2006, Stokes Research Institute-University of Limerick, Limerick, Ireland. PU 05) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr., "Spreading Characteristics and Microscale Evaporative Heat Transfer in an Ultra-Thin Film Containing a Binary Mixture," ASME Journal of Heat Transfer, 2006, 128, pp. 1266- 1275. PU 06) Peter C. Wayner, Jr., "Effects of Interfacial Phenomena and Conduction on an Evaporating Meniscus," Paper # ICNMM2007-30021 Presented at Fifth International Conference on Nanochannels, Microchannels and Minichannels, June 18-20, 2007. Universidad De Las Americas Puebla, Mexico. PU 07) Sashidhar S. Panchamgam, Joel L. Plawsky, and Peter C. Wayner, Jr. , "Experimental Evaluation of Marangoni Shear in the Contact Line Region of an Evaporating 99 + % Pure Octane Meniscus." ASME Journal of Heat Transfer, 129, pp. 1476-1485, 2007. PU 08) Paper # 2008-0818. Proceedings of the 22nd Symposium on Gravity-Related Phenomena in Space Exploration, January 7-10, 2008, Reno, NV. "R&D Work on the Constrained Vapor Bubble System for a Microgravity Experiment", Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. PU-09) Arya Chatterjee, Joel L. Plawsky and Peter C. Wayner, Jr. " Continuum Models with Slip for the Evaporating Meniscus and Comparison with Experimental Results" Paper Number: ICNMM2008-62080. Sixth International Conference on Nanochannels, Microchannels and Minichannels,", TU Darmstadt, Germany, June 23-25, 2008. PU-10) Sashidhar S. Panchamgam, Arya Chatterjee, Joel L. Plawsky, and Peter C. Wayner, Jr. , "Comprehensive experimental and theoretical study of fluid flow and heat transfer in a microscopic evaporating meniscus in a miniature heat exchanger." International Journal of Heat and Mass Transfer, 51, pp. 5368-5379, 2008. PU-11) "Review of the Effects of Surface Topography, Surface Chemistry, and Fluid Physics on Evaporation at the ContactLine", Plawsky, Joel; Ojha, Manas; Chatterjee, Arya, and Wayner, Jr. Peter C.,Chemical Engineering Communications 196, no. 5 (2009): 658-696. Other products of our work: THESES [T] T 1) Sashidhar S. Panchamgam, "Interfacial Phenomena and Microscale Transport Processes in Evaporating Ultrathin Menisci" Ph. D. Thesis, The Isermann Department of Chemical and Biological Engineering, Rensselaer Polytechnic Institute, Troy, NY, September, 2006. T2) Manas Ojha, "Contact Line Wettability and Transport During Phase Change Processes From Nanostructured Surfaces" Ph. D. Thesis, The Isermann Department of Chemical and Biological Engineering, Rensselaer Polytechnic Institute, Troy, NY, August, 2008. ANNUAL PROGRESS REPORTS [ANN]: ANN 1) Peter C. Wayner, Jr and Joel L. Plawsky, "R & D Work on the Constrained Vapor Bubble System for a Microgravity Experiment: First Annual Progress Report for NNC05GA27G for the Period April 5, 2005 –February 4, 2006" submitted February, 2006. ANN 2) Peter C. Wayner, Jr and Joel L. Plawsky, "R & D Work on the Constrained Vapor Bubble System for a Microgravity Experiment: Second Annual Progress Report for NNC05GA27G for the Period April 5, 2006 –February 4, 2007" submitted February, 2007. ANN 3) Peter C. Wayner, Jr and Joel L. Plawsky, "R & D Work on the Constrained Vapor Bubble System for a Microgravity Experiment: Third Annual Progress Report for NNC05GA27G for the Period April 5, 2007 –April 4, 2008" submitted February, 2008. ANN 4) Peter C. Wayner, Jr and Joel L. Plawsky, "R & D Work on the Constrained Vapor Bubble System for a Microgravity Experiment: Fourth Annual Progress Report for NNC05GA27G for the Period April 5, 2008 –April 4, 2009" submitted February, 2009. FINAL REPORT R & D WORK ON THE CONSTRAINED VAPOR BUBBLE SYSTEM FOR A MICROGRAVITY EXPERIMENT SUMMARY OF RESEARCH FOR NNC05GA27G For the period April 5, 2005 – June 1, 2009. Peter C. Wayner, Jr and Joel L. Plawsky, Report Date: June 22, 2009
Bibliography Type:	Description: (Last Updated: 09/25/2012)
Articles in Peer-reviewed Journals	Plawsky JL, Ojha M, Chatterjee A, Wayner PC Jr. "Review of the effects of surface topography, surface chemistry, and fluid physics on evaporation at the contact line." Chemical Engineering Communications, 2009 May;196(5): 658-96. http://dx.doi.org/10.1080/00986440802569679 , May-2009
Articles in Peer-reviewed Journals	Panchamgam SS, Chatterjee A, Plawsky JL, Wayner PC Jr. "Comprehensive experimental and theoretical study of fluid flow and heat transfer in a microscopic evaporating meniscus in a miniature heat exchanger." International Journal of Heat and Mass Transfer. 2008 Oct;51(21-22):5368-79. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2008.03.023 , Oct-2008

Articles in Peer-reviewed Journals	Panchamgam SS, Plawsky JL, Wayner PC Jr. "Experimental evaluation of Marangoni shear in the contact line region of an evaporating 99+% pure octane meniscus." Journal of Heat Transfer. 2007 Nov;129(11):1476 (10 pages). http://dx.doi.org/10.1115/1.2759970 , Nov-2007
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Articles in Peer-reviewed Journals	Panchamgam SS, Plawsky JL, Wayner PC Jr. "Microscale heat transfer in an evaporating moving extended meniscus." Experimental Thermal and Fluid Science. 2006 Aug;30(8):745-54. http://dx.doi.org/10.1016/j.expthermflusci.2006.03.004 , Aug-2006
Articles in Peer-reviewed Journals	Gokhale SJ, Plawsky JL, Wayner PC Jr. "Spreading, evaporation, and contact line dynamics of surfactant-laden microdrops." Langmuir. 2005 Aug 2;21(18):8188-97. http://dx.doi.org/10.1021/la050603u , Aug-2005
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